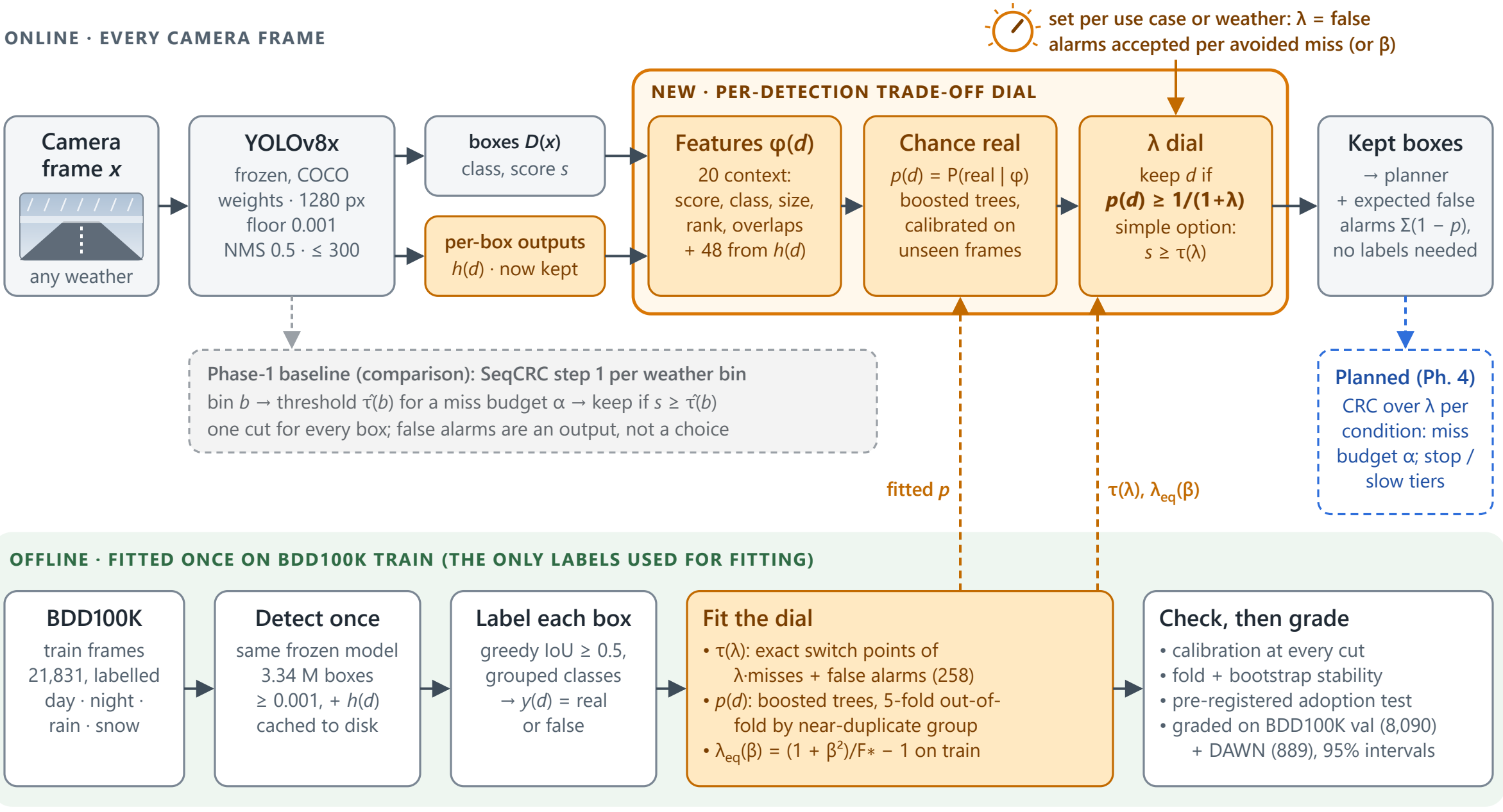

ONLINE · EVERY CAMERA FRAME
set per use case or weather: λ = false alarms accepted per avoided miss (or β)
NEW · PER-DETECTION TRADE-OFF DIAL
Camera frame x
any weather
YOLOv8x
frozen, COCO weights · 1280 px
floor 0.001
NMS 0.5 · ≤ 300
boxes D(x)
class, score s
per-box outputs
h(d) · now kept
Features φ(d)
20 context: score, class, size, rank, overlaps + 48 from h(d)
Chance real
p(d) = P(real | φ)
boosted trees, calibrated on unseen frames
λ dial
keep d if p(d) ≥ 1/(1+λ)
simple option: s ≥ τ(λ)
Kept boxes
→ planner + expected false alarms Σ(1 − p), no labels needed
Phase-1 baseline (comparison): SeqCRC step 1 per weather bin
bin b → threshold τ̂(b) for a miss budget α → keep if s ≥ τ̂(b)
one cut for every box; false alarms are an output, not a choice
Planned (Ph. 4)
CRC over λ per condition: miss budget α; stop / slow tiers
fitted p
τ(λ), λeq(β)
OFFLINE · FITTED ONCE ON BDD100K TRAIN (THE ONLY LABELS USED FOR FITTING)
BDD100K
train frames 21,831, labelled day · night · rain · snow
Detect once
same frozen model 3.34 M boxes ≥ 0.001, + h(d) cached to disk
Label each box
greedy IoU ≥ 0.5, grouped classes → y(d) = real or false
Fit the dial
• τ(λ): exact switch points of λ·misses + false alarms (258)
• p(d): boosted trees, 5-fold out-of-fold by near-duplicate group
• λeq(β) = (1 + β²)/F* − 1 on train
Check, then grade
• calibration at every cut
• fold + bootstrap stability
• pre-registered adoption test
• graded on BDD100K val (8,090) + DAWN (889), 95% intervals
unchanged (as in the Phase-1 baseline)
new in our method
planned for Phase 4
baseline path (comparison)
offline fitting